

CURRICULUM VITAE

MAHENDRA PRATAP YADAV

Assistant Professor

**Department of Computer Science & Engineering
Indian Institute of Information Technology, Pune,
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OBJECTIVE:

Looking forward with my organization where there is an opportunity to share, contribute and upgrade my knowledge for development of self and the organization.

CURRENT POSITION:

- Presently working as an Assistant Professor in Department of Computer Science and Engineering at **Indian Institute of Information Technology Pune (IIIT Pune), Maharashtra – 411041 (01 December-2022 to till).**

AREA OF RESEARCH:

- **Cloud Computing, Artificial Intelligence Machine Learning, Formal Method**

PROFESSIONAL QUALIFICATION:

- **Ph. D.**
Thesis Title: Optimal Resource Provisioning for Containerized Applications
Specialization: Cloud Computing
Department: Computer Science and Engineering
Institute Name: Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India
Registration Date: 12-01-2016
Award Date: 09-09-2021
- **M. Tech.**
Thesis Title: An Interface Formalism of Web Service Based Protocol Specification and Component Adapter Using Pi-Calculus.
Specialization: Information Security
Department: Computer Science and Engineering
Institute Name: Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India
CPI: 8.125 (81.20%)
Session: 2010-2012
- **B. Tech. in Information Technology** from U.P. Technical University Lucknow (U.P) with 67.04% in the session: 2002-2006.

OTHER QUALIFICATION:

- Qualified **UGC-NET-2018**
- Qualified **GATE-2008**

ACADEMIC PROFILE:

- Intermediate passed in 1999 from U.P. Board Allahabad (U.P.).
- High School passed in 1997 from U.P. Board Allahabad (U.P.).

TEACHING EXPERIENCE: (Total-06 Years)

- Worked as an Assistant Professor in the department of Computing Science & Engineering at SRM University, NCR Campus Modinagar, Ghaziabad, UP – 201204 **(01 July-2022 to 30-11-2022)**.
- Worked as an Assistant Professor (Grade-II) in School of Computing Science & Engineering at VIT University Bhopal, Madhya Pradesh – 466114 **(02-August-2021 to 17-June-2022)**.
- Worked as a Guest faculty in the Department of Computer Science and Engineering at Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India **(30-July-2015 to 16-December-2015)**.
- Worked as a Visiting faculty in the Department of Computer Science and Engineering at Motilal Nehru National Institute of Technology Allahabad, Prayagraj, India **(21-January-2015 to 31-May-2015)**.
- Worked as Assistant Professor in the Department of Computer Science and Engineering at College of Engineering & Rural Technology Meerut (U.P.) India **(28-August-2012 to 30- April-2014)**.
- Worked as Lecturer in the Department of Computer Science and Engineering at College of Engg. & Rural Technology Meerut (U.P.), India **(21-August-2007 to 24- July-2010)**.
- Worked as Lecturer in the Department of Information Technology at Bhagwant Institute of Technology Muzaffarnagar (U.P.), India **(01-February-2007 to 20- August-2007)**.

LIST OF ADMINISTRATIVE POSITIONS HELD:

- Faculty In-charge Internet and Networking (From October 2025 to Continue)
- Chairperson (Physical Stock Verification Committee) (Financial Year 2023-24)
- Faculty In charge (National Education Policy (NEP-2020)) (From 09th February 2024 to Continue)
- Warden- Boys Hostel (From 03rd October 2023 to 02 September 2024)
- Nodal Officer Visvesvaraya PHD Scheme (From 13th September 2023 to September-2025)
- Faculty In-charge Hindi Rajbhasha Committee (From 02nd June 2023 to Continue)

Professional Academic Contributions:

- **Board of Studies (BoS) Member**, Department of Information Technology, Poornima College of Engineering, Jaipur.
- **Board of Studies (BoS) Member**, Department of Computer Science & Engineering (Cyber Security), School of Engineering (SoE), Dayananda Sagar University.
- **Syllabus Auditor** Department of Computer Engineering, Pimpri Chinchwad College of Engineering, Pune,
- **Board of Studies (BoS) Member**, Pune Vidyarthi Griha's College of Engineering & S. S. Dhamankar Institute of Management, Nashik

AREA OF INTREST:

- Computer Programming, Data structure, Theory of Automaton, Design analysis of algorithm

SOFTWARE SKILLS:

- **LANGUAGES:** C, C++, Java.
- **DATABASE:** SQL.

RESEARCH PAPERS (COMMUNICATED /ACCEPTED/ PUBLISHED) IN JOURNALS (SCIE):

- Subhash Yadav, Shubham Shukla, Ritu Tiwari, and **Mahendra Pratap Yadav**. Adaptive path planning in complex environments for mobile robots via fusion of reinforcement learning and hybrid swarm intelligence. Intel Serv Robotics 19, 84 (2026). <https://doi.org/10.1007/s11370-026-00735-0>
- V. Gupta, N. Jain, A. Anand, D. Jain, M. P. Yadav **Mahendra Pratap Yadav** and Y. P. Singh, "LogHyper-IB: Spatio-Temporal Hypergraph Representation Learning via Information Bottleneck for Anomaly Detection in Distributed Systems," in IEEE Access, vol. 14, pp. 60629-60645, 2026, doi: 10.1109/ACCESS.2026.3682603.
- Subhash Yadav, Shubham Shukla, Ritu Tiwari, and **Mahendra Pratap Yadav**, "Path planning for mobile robots with waypoint management using a hybrid nature inspired optimization algorithm" published in Cluster Computing (2025) DOI-<https://doi.org/10.1007/s10586-025-05111-8>
- Padma Nimbhore, Tanmoy Hazra, Ritu Tiwari, and **Mahendra Pratap Yadav**, "Classification of Cotton Crop Disease using Hybrid Model and MDfC feature extraction method", accepted for publication in the Journal of Phytopathology.
- Sanjeev Sharma, **Mahendra Pratap Yadav**, "A collective approach to reach known and unknown target in multi agent environment using nature inspired algorithms", published in Cluster Computing (2024). <https://doi.org/10.1007/s10586-024-04523-2>
- Rupesh Kumar Dewang, **Mahendra Pratap Yadav**, Surbhit Awasthia, Om Raja, Arvind Mewadaa,, Kamlakant Laxman Bawankulea "Data Secure Application: An Application that Allows Developers to Store User Data Securely using Blockchain and IPFS" published in Multimedia Tools and Applications Volume 83, pages 45491–45517 (2023).
- Brajesh Kumar Umrao, **Mahendra Pratap Yadav** and Dharmendra Kumar Yadav. "Adaptive model for the placement of network services" published in Journal of Information Science and Engineering 40, DOI: 10.6688/JISE.202401_40 (1).0010 ().

- Nisha Pal, **Mahendra Pratap Yadav** and Dharmendra Kumar Yadav, “Appropriate Number of Analogues in Analogy Based Software Effort Estimation using Quality Datasets” published in Journal of Cluster Computing, publisher Springer, (Indexing - SCIE).
- **Mahendra Pratap Yadav**, Rohit and Dharmendra Kumar Yadav, “**Resource Provisioning Through Machine Learning in Cloud Services**” published 24 July 2021 in Arabian Journal for Science and Engineering Vol. No 47, 1483–1505 (2022) (ISSN: 2191-4281) publisher Springer, <https://doi.org/10.1007/s13369-021-05864-5>, (Indexing - SCIE).
- Mahendra Pratap Yadav, Rohit and Dharmendra Kumar Yadav, “**Maintaining the Container Sustainability through Machine Learning**” published 22 July 2021 in Journal of Cluster Computing (2021) Vol. No.- 24(4) page no- 3725–3750, publisher Springer, <https://doi.org/10.1007/s10586-021-03359-4>, ISSN: 1573-7543, (Indexing - SCIE).
- **Mahendra Pratap Yadav**, Nisha Pal and Dharmendra Kumar Yadav, “**A Formal Approach for Docker Container Deployment**” published 25 May 2021, in Journal of Concurrency and Computation: Practice and Experience (2021) publisher Wiley, Volume 33, Issue 20, <https://doi.org/10.1002/cpe.6364>, ISSN:1532-0634, (Indexing - SCIE).
- **Mahendra Pratap Yadav**, Nisha Pal and Dharmendra Kumar Yadav “**Resource Provisioning for Containerized Applications**”, published 13 May 2021 in Journal of Cluster Computing(2021) Vol. No.- 24(4) page no-2819–2840, publisher Springer <https://doi.org/10.1007/s10586-021-03293-5>, ISSN: 1573-7543, (Indexing - SCIE).

RESEARCH PAPERS (COMMUNICATED /ACCEPTED/ PUBLISHED) IN JOURNALS (SCOPUS):

- Sharma, Alesh Kumar, Ritu Tiwari, Rahul Dixit, and **Mahendra Pratap Yadav**. "Modelling of features of fusion using a hybrid swarm optimization algorithm with deep learning methodology for copy-move image forgery detection." International Journal of System Assurance Engineering and Management (2025): 1-21.
- **Mahendra Pratap Yadav**, Nisha Pal and Dharmendra Kumar Yadav, “**Verification of Cloud System Elasticity using BigMC**” published in Journal of System Assurance Engineering and Management, publisher Springer Vol. 13, Page No. 2208–2220 (2022). <https://doi.org/10.1007/s13198-022-01627-z> (Indexing -Scopus).
- Nidhi Sindhwani, Rohit Anand, Meivel S., Rati Shukla, **Mahendra Pratap Yadav**, Vikash Yadav, “**Performance Analysis of Deep Neural Networks Using Computer Vision**”, published 14-October 2021 with Volume 8 and Issue 29 in EAI Endorsed Transactions on Industrial Networks and Intelligent Systems, Europe (ISSN: 2410-0218), <http://dx.doi.org/10.4108/eai.13-10-2021.171318>, (Indexing - Scopus).
- **Mahendra Pratap Yadav**, Gaurav Raj, Harishchandra Akarte, and Dharmendra Kumar Yadav, “**Horizontal Scaling for Containerized Application Using Hybrid Approach**” published 31-12-2020 in International journal of Ingénierie des Systèmes d’Information, Vol. 25, No. 6, pp. 709-718. <https://doi.org/10.18280/isi.250601>, ISSN: 1633-1311 (print); 2116-7125 (online) (Indexing - Scopus).
- **Mahendra Pratap Yadav**, Harishchandra Akarte, and Dharmendra Kumar Yadav, “**Container Elasticity: Based on Response Time using Docker**” published 12 October, 2020, in International journal of Recent Advances in Computer Science and Communications. (Formerly: Recent Patents on Computer Science) publisher Bentham science, ISSN (Online): 2666-2566, pp-1-13 doi:10.2174/2666255813999201012192010, (Indexing - Scopus).
- **Mahendra Pratap Yadav**, Akanksha Kunwar, Ram Chandra Bhushan, Dharmendra Kumar Yadav, “**Web Server Workload Prediction using Time Series Model**” accepted in International Journal of

Advanced Intelligence Paradigms publisher Inderscience-2020, ISSN online 1755-0394, DOI: 10.1504/ijaip.2022.10034175 (Indexing- Scopus).

PATENT (GRANT/PUBLISHED/COMMUNICATED):

- Design Patent granted on the topic “**AI-Based Energy Consumption Monitoring Device**” in 2025.
- Design Patent granted on the topic “**Blockchain-Based Medical Data Storage Cloud Device**” in 2025.
- Patent publishing on the topic “**Mirror List Query Architecture of Multi Architecture Cloud Platform Mirror Warehouse**” in 2022.

RESEARCH PAPERS (PUBLISHED/ACCEPTED) IN BOOK CHAPTERS:

- Anand, Anand, **Mahendra Pratap Yadav**, Anita Yadav, and R. Kumar, (2026). Self-Corrective Retrieval Augmented Generation System for Conversational Text. In F. Dehghan (Ed.), Harnessing AI-Generated Content in Language Education (pp. 189-212). IGI Global Scientific Publishing. <https://doi.org/10.4018/979-8-3373-2721-1.ch008>
- **Mahendra Pratap Yadav**, Harishchandra Akarte, Dharmendra Kumar Yadav, “**Efficient Resources Utilization of Containerized Applications using TOPSIS**” published as a book chapter in book with entitled Recent Developments in Artificial Intelligence & Communication Technologies, publisher Bentham Science in 2022 ISSN **978-1-68108-967-6**.
- **Mahendra Pratap Yadav**, Harishchandra Akarte, Dharmendra Kumar Yadav, “**An Approach for Load Balancing Through Genetic Algorithm**” published as a book chapter in book with entitled “Smart and Sustainable Intelligent Systems” publisher Scrivener Publishing (Wiley) in March 2021, pp- 511-524, ISBN: 978-1-119-75058-1, <https://doi.org/10.1002/9781119752134.ch36>.

RESEARCH PAPERS (PUBLISHED/ACCEPTED) IN CONFERENCES:

- Aman Anand; **Mahendra Pratap Yadav** “A Levy-Flight Enhanced Whale Optimization Algorithm for Load Balancing in Edge IoT” accepted in the 2026 IEEE International Conference on AI Engineering and Innovation (AIEI 2026) to organize at NIT, Jamshedpur during 26 – 28 March, 2026
- Jahnvi Tiwari, Praveen raj, Anushka Garg, **Mahendra Pratap Yadav**, Priyanka Singh, Dheeraj Dubey “A MERN Stack Framework for Real Estate Platforms Leveraging Conversational AI and Predictive Analytics” accepted in the 2025 IEEE 1st International Conference on Recent Trends in Computing and Smart Mobility (RCSM) to organize at MANIT, bhopal during 26 – 28 March, 2026
- Aman Anand; **Mahendra Pratap Yadav** “A Whale Optimization Algorithm for Resource Optimized Load Balancing in IoT Based Edge Network” 2025 International Conference on Artificial intelligence and Emerging Technologies (ICAIET), Bhubaneswar, India, 2025, pp. 1-6, doi: 10.1109/ICAIET65052.2025.11211078.
- Nisha Pal, **Mahendra Pratap Yadav** and Dharmendra Kumar Yadav, “**Modeling and Verification of Web Services Composition Using CWB-NC Tool**” published (22 June 2021) in the “INCET 2021:2nd International Conference of Emerging Technologies” ISBN:978-1-7281-7029-9 organize on date 21th -

23th May, 2021 at Jain College of Engineering & Technology, Belgaum Karnataka, INDIA.

- **Mahendra Pratap Yadav**, Nisha Pal, and Dharmendra Kumar Yadav, “**Workload Prediction over Cloud Server using Time Series Data**” published (15 march, 2021) in the "Confluence-2021:11th International Conference on Cloud Computing, Data Science & Engineering (28th - 29th January, 2021), ISBN:978-1-6654-1451-7 at Amity University Uttar Pradesh, Noida, India.
- **Mahendra Pratap Yadav**, and Dharmendra Kumar Yadav “**Workload Prediction for Cloud Resource Provisioning using Time Series Data**”, presented in 10th International Conference on Soft Computing for Problem Solving - SocProS 2020. Springer, Singapore, December 18-20, 2020, Organized by Indian Institute of Technology Indore, 14 October 2021, vol 139, pages 447-459, ISBN- 978-981-16-2712-5.
- **Mahendra Pratap Yadav**, Mhd Feeroz, Vinod Kumar Yadav, “**Mining the Customer Behavior using Web Usage Mining in E-commerce**”, published (31 December 2012) in (ICCCNT-2012) Third International Conference on Computing Communication and Networking Technology, July 26 -28, 2012, INSPEC Accession Number: 13252164, organized by CCET Tamilnadu, India.
- Vinod Kumar Yadav, **Mahendra Pratap Yadav**, and Dharmendra Kumar Yadav, “**Reliable Task Allocation in Heterogeneous Distributed System with Random Node Failure: Load Sharing Approach**”, published (24 December 2012) in TURING100 - INTERNATIONAL CONFERENCE ON COMPUTING SCIENCES (ICCS-2012), 14-15 September-2012, ISBN- 978-1-4673-2647-6, organized by Lovely Professional University Punjab, India.
- **Mahendra Pratap Yadav**; Pankaj Keserwani; Shefali Gosh Samundar; “**An Efficient Web Mining Algorithm for Web Log Analysis: E-Web Miner**”, , published (07 May 2012) in first International Conference on Recent Advance in Information Technology, 15-17 March 2012, ISBN:978-1-4577- 0697-4, Organized by IIT (ISM) Dhanbad, India.
- C.P. Sahu,; **Mahendra Pratap Yadav**.; Manoj Rana.; Ritesh Srivastava; “**A Simple Delayed Acknowledgement Scheme**”, published (18 November 2010) in International Conference on Computer and Communication Technology (ICCCT), 17-19 Sept. 2010 Digital Object Identifier: 10.1109/ICCCT.2010.5640509 Publication Year: 2010 , Page(s): 393 – 396, ISBN:978-1-4244-9034-9, organized by MNNIT Allahabad, India.
- **Mahendra Pratap Yadav**, Mhd Feeroz, and D. K, Yadav “**Mining the Customer Behavior using asociation rule in E-commerce**” published in National conference Jigyasa 2012 Organized by BIT M. Nagar(U.P.).

KEYNOTE SPEAKER/ SESSION CHAIRED:

- Delivered a talk on “**DevOps: Software Development and Operations**” at Dr. Babasaheb Ambedkar Technological University, Lonere, Maharashtra (2023).
- Delivered a talk on “**Artificial Intelligence and it’s Real Life Applications**” at ISBM College of Engineering, Pune Maharashtra (2023).
- Session the chaired in “**International Conference On Intelligent Systems and Smart Infrastructure (ICISSI-2022)**” which is jointly organized by Shambhunath Institute of Engineering and Technology, Prayagraj (U.P.) INDIA, Institute Of Engineering and Technology, Lucknow (U.P.) INDIA and Manipal University Jaipur, India
- Delivered a talk (as a Keynote speaker) on “**Container Based Virtualization: A Modern Approach to Deliver the Cloud Services**” in 16th International Conference for Internet Technology and Secured Transactions (ICITST-2021).

CONFERENCES ATTENDED:

- Attended a tutorial on “**Network Security and WiMAX**” in 2010 IEEE International Conference on Computer and Communication Technology (ICCCT), 2010 organized by MNNIT Allahabad.
- Attended a tutorial on “**Cloud Computing and security**” in 2011 IEEE International Conference on Computer and Communication Technology (ICCCT), 2011 organized by MNNIT Allahabad.

FDPs/ WORKSHOPS/SEMINARS ATTENDED:

- One week **AICTE Training and Learning (ATL) Faculty Development Program** on “**Artificial Intelligence**”, from 10-12-2019 to 14-12-2019, at MNNIT Allahabad, India.
- One week **Workshop** on “**Current Trends in Cyber Security (CTC-2018)**”, from 27-01-2018 to 31-01-2018, organized by department of computer science and engineering at MNNIT Allahabad, India.
- One week **Faculty Development Program** on “**Cloud Computing with AWS**” from 07-08-2017 to 13-08-2017, organized by E&ICT Academy, IIT Guwahati held at KKR & KSR Institute of Technology & Science, Guntur, Andhra Pradesh, India.
- One week **Workshop** on “**Data Analytics and Machine Learning with R**” from 29-05-2017 to 02-06-2017, organized by department of computer science and engineering at MNNIT Allahabad, India.
- One week **Workshop** on “**Network Simulation (WNS-2016)**” from 28-09-2016 to 02-10-2016, organized by department of computer science and engineering at MNNIT Allahabad, India.
- Two weeks (One week online (from 27-04-2015 to 24-05-2015) and one week physical participation (from 25-05-2015 to 30-05-2015)) **ISTE ITTP Workshop** on “**Introduction to Design of Algorithm**” conducted by IIT Kharagpur at remote center MNNIT Allahabad, India.
- Two weeks **Faculty Development Program** on “**Curriculum Enhancement Awareness in Information in Information Security**” (**SOOCHNA SUREKSHA SAMIKSHA-2009**) from 14-12-2009 to 24-12-2009, organized by department of computer science and engineering at MNNIT Allahabad, India and Sponsored by AICTE.

STRENGTHS:

- Good communication and interpersonal skills.
- Ability to work with co-operation.
- Leadership quality

Declaration:

I hereby declare that the information furnished above is true to the best of my knowledge.

Place:

Date:

(MAHENDRA PRATAP YADAV)

Note*: Relevant documents will be produced on demand.